STAFF REVIEW · H446 257 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It can apply similar computations to many data elements in parallel.

2. It decodes and coordinates instruction execution.

3. Boolean.

4. 4×8=32 bits=4 bytes. Unicode contains many more characters, and UTF encodings may use more than one byte for a character; 8-bit storage here is an assumption.

I Do — source worked steps

1. Predict first — A proximity or motion sensor detects someone approaching and passes a signal to the controller.

2. Explain and check — A motor and driver create movement; safety sensing reduces the risk of closing onto a person.

We Do — guided CHECK

1. Predict first
A push button is a clear intentional input and the controller processes it under safe traffic timing rules.
Explain and check
Outputs include signal lights plus appropriate audible or tactile cues to improve accessibility.

You Do — source mark schemes

1. [4] Temperature sensor measures temperature [2]; display or data logger presents/stores readings [2].

2. [6] Temperature sensor [1] measures physical quantity [1]; ADC where needed [1]; controller compares with threshold [1]; motor actuator [1] opens vent [1].

3. [6] Button input should be accessible [1]; visual indications [1]; audible signal [1]; tactile rotating cone/vibration [1]; suitable signal timings [1]; prevent misleading or unsafe outputs [1].

Exit Ticket — source answer

1. False. It is an output actuator.